

Climate Change and Sustainability

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Learning Objectives

1. Explain why climate change is important to Canada
2. Understand core economic concepts related to climate change and sustainability:
 - a Define and explain the concepts of externalities, public goods, and market failures as they relate to environmental economics
 - b Identify how these concepts influence climate change policies and practices

Definitions

Climate Change

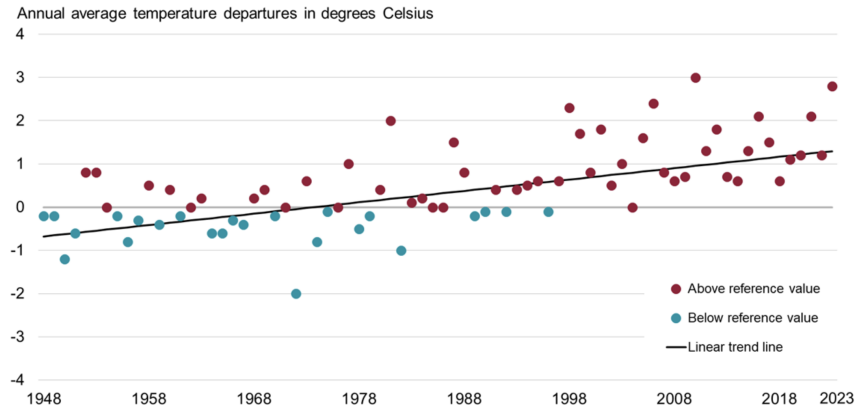
- ▶ long-term changes in average weather patterns, mainly driven by human activities such as the burning of fossil fuels, deforestation, and industrial processes
- ▶ increasing concentration of greenhouse gases (GHGs) like carbon dioxide (CO₂) and methane (CH₄) traps heat in the Earth's atmosphere, leading to global warming and severe weather events

Sustainability

- ▶ meeting the needs of the present without compromising the ability of future generations to meet their own needs
- ▶ involves a balance between economic growth, environmental preservation, and social well-being

Why is climate change important to Canada?

Annual average temperature departures from the 1961 to 1990 reference value, Canada, 1948 to 2023



Why is climate change important to Canada?

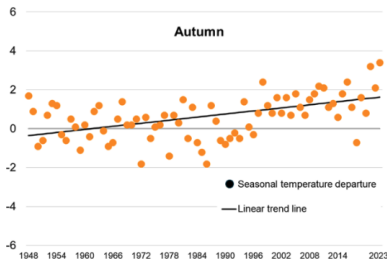
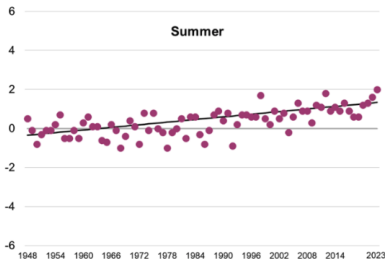
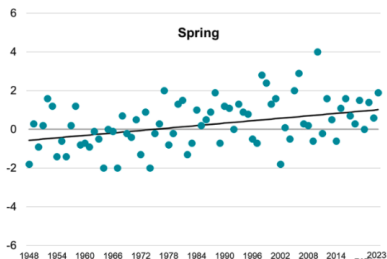
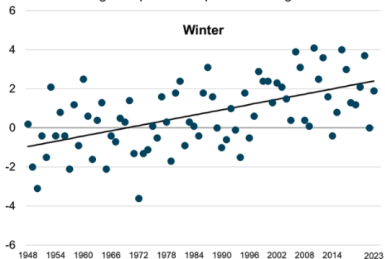
Canada's annual temperature change

- ▶ average temperature for the year 2023 was 2.8 degrees Celsius ($^{\circ}\text{C}$) above the 1961-1990 reference value
 - ▶ making it second warmest year since 1948
- ▶ From 1948 to 2023, there is a trend in annual average temperature departures, showing 2.0 $^{\circ}\text{C}$ of warming over that period
- ▶ annual average temperatures were consistently above or equal to the reference value from 1997 onward

Why is climate change important to Canada?

Seasonal average temperature departures compared with the 1961 to 1990 reference value, Canada, 1948 to 2023

Seasonal average temperature departures in degrees Celsius



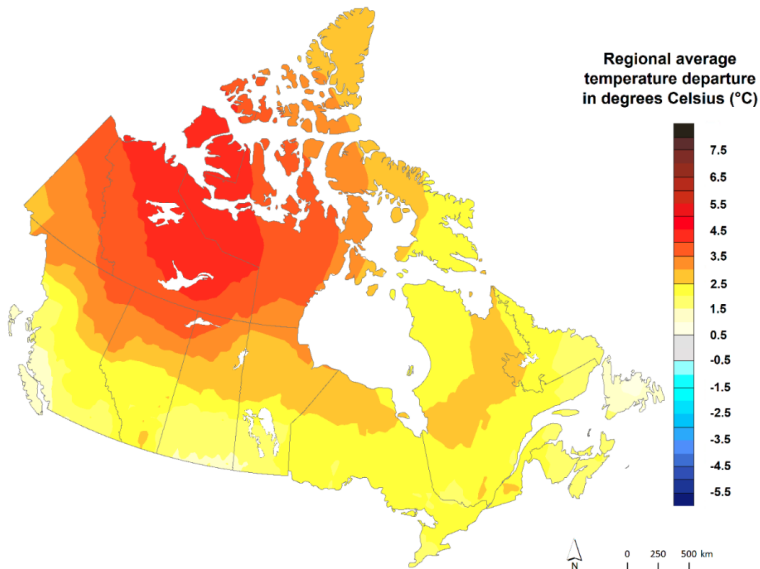
Why is climate change important to Canada?

Canada's seasonal temperature change

- ▶ warming trends were detected for all four seasons:
 - ▶ winter, with an increase of 3.4°C
 - ▶ spring, with an increase of 1.9°C
 - ▶ summer with an increase of 1.9°C
 - ▶ Fall, with an increase of 2.0°C
- ▶ the warmest winter and spring recorded were in 2010
- ▶ the warmest summer and fall were in 2023

Why is climate change important to Canada?

Regional average temperature departures from the 1961 to 1990 reference value, Canada, 2023



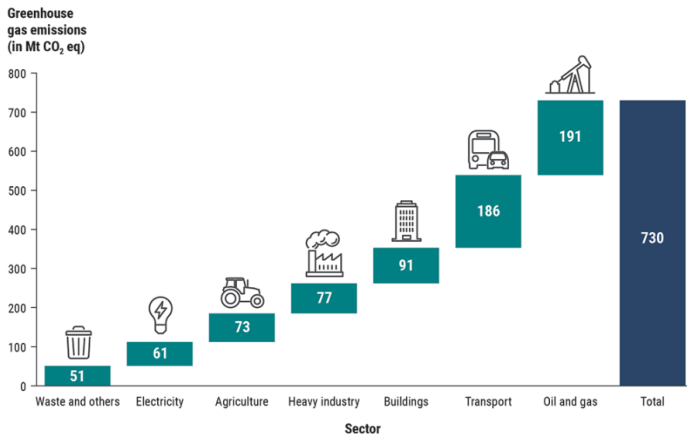
Why is climate change important to Canada?

Regional temperature departures

- ▶ All regions of Canada experienced annual temperatures above the 1961-1990 reference value
- ▶ Most of northern Canada and the northern parts of British Columbia, the Prairies, Ontario and Quebec, experienced temperatures significantly above the baseline average
- ▶ Eastern Newfoundland and Labrador and western British Columbia experienced the least annual temperature departure from the baseline average

Canada's economy depends on emission-intensive sectors

Oil and gas production accounts for the largest share of Canada's greenhouse gas emissions by economic sector (2019)



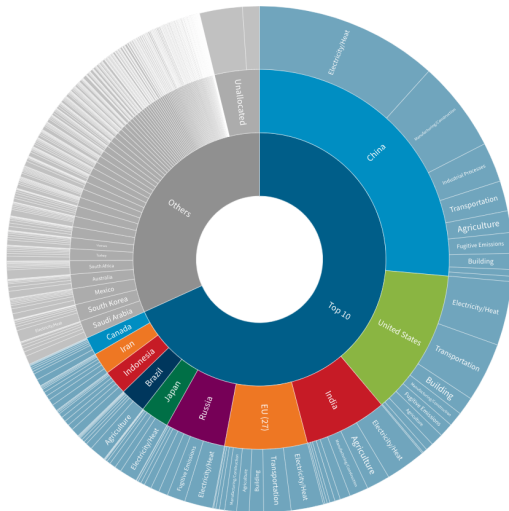
Source: Based on emission data from Canada's 2021 National Inventory Report

Canada's economy depends on emission-intensive sectors

In 2019,

- ▶ Canada emitted 730 megatonnes of carbon dioxide equivalent
- ▶ oil and gas production accounted for the largest share of Canada's greenhouse gas emissions by economic sector

Comparing Canada's GHG emissions to global levels



Source: Global GHG Emissions 2019 excluding LUCF. [Climate Watch](#) • The EU 27 is considered a country.

*Bunker fuels include international aviation and shipping that are not included in country totals. Other territories include regions not covered by Climate Watch country data. See Climate Watch for country level land-use change and forestry and bunker fuel emissions.

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Comparing Canada's GHG emissions to global levels

- ▶ China is the biggest emitter → 26.4% of global greenhouse gas emissions
- ▶ United States → 12.5%
- ▶ India → 7.06%
- ▶ European Union → 7.03%

Canada contributes approximately 1.5% of the global GHG emissions.

The Economics of Climate Change

Externalities and Market Failures:

- ▶ **Negative Externalities:** climate change is a classic negative externality, where the social costs of emissions are not reflected in market prices
- ▶ Carbon emissions from transportation, energy production, and industry impose costs on society, including:
 - ▶ **health costs** due to air pollution
 - ▶ **Environmental damage** like rising sea levels and more frequent extreme weather events
 - ▶ **Economics losses** in sectors like agriculture, tourism, and infrastructure

Example:

- ▶ Canada's oil sands present a case of over-reliance on carbon-intensive energy, where the social cost of pollution is externalized

The Economics of Climate Change

Public Goods:

- ▶ non-excludable and non-rivalrous resources from which everyone benefits
- ▶ examples: clean air, stable climates, and healthy ecosystems
- ▶ since no one can be excluded from enjoying a stable climate, there is little incentive for individual actors (countries, companies, or individuals) to voluntarily reduce emissions, leading to the **free rider problem**
- ▶ Canada, contributing only 1.5% of global emissions, is susceptible to the global free-rider problem, depending on international cooperation to combat climate change

Policy Responses to Climate Change: Carbon Pricing as a Market-Based Solution

Pigovian Taxes:

- ▶ carbon taxes are an example of Pigovian taxes
- ▶ designed to correct a negative externality by making polluters pay the social cost of their emissions
- ▶ by internalizing the cost of carbon, carbon pricing aligns private costs with social costs

Canada's national carbon pricing policy:

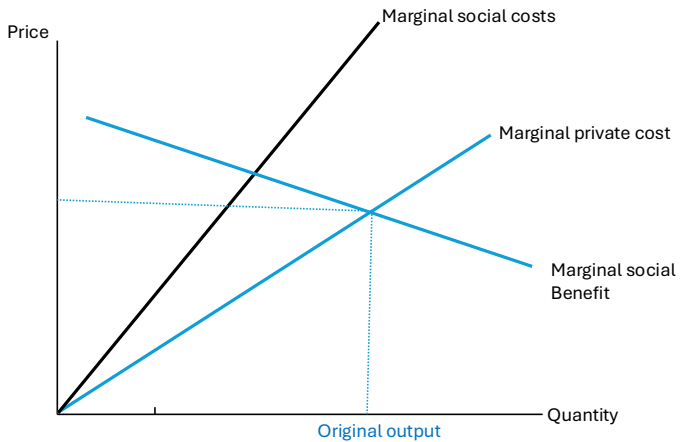
- ▶ implemented in 2019
- ▶ is a model for a Pigovian tax in action
- ▶ targeting carbon price of *CAD*\$170 per tonne by 2030
- ▶ encourages reductions in emissions by making carbon-intensive goods and services more expensive

Policy Responses to Climate Change: Carbon Pricing as a Market-Based Solution

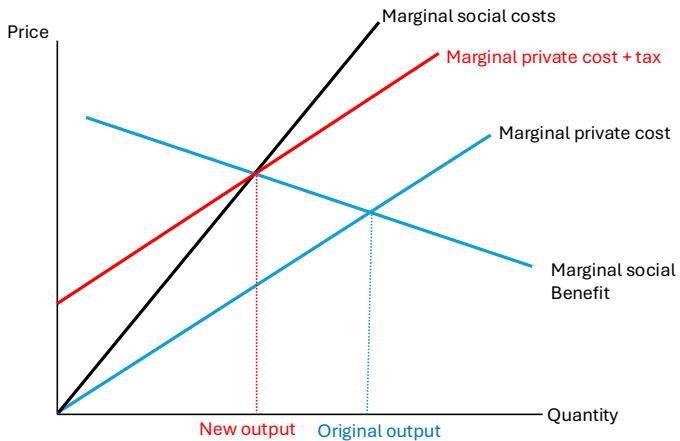
British Columbia's Carbon Tax:

- ▶ introduced in 2008
- ▶ starting at *CAD*\$10 per tonne of CO₂ and gradually increasing
- ▶ led to a significant reduction in emissions without harming the economy
- ▶ proving the effectiveness of Pigovian taxes in correcting market failures

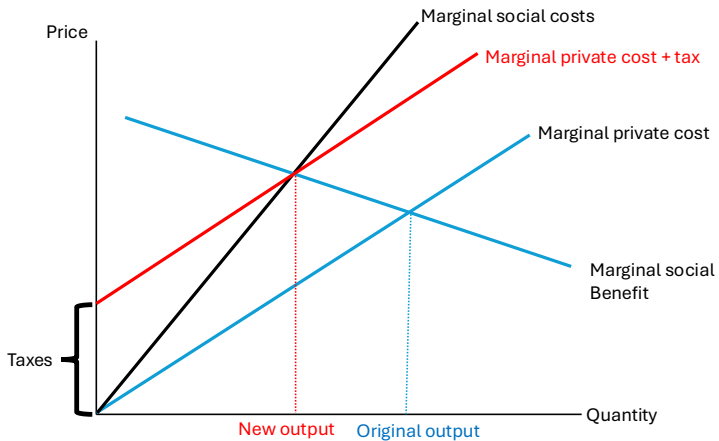
Carbon Tax



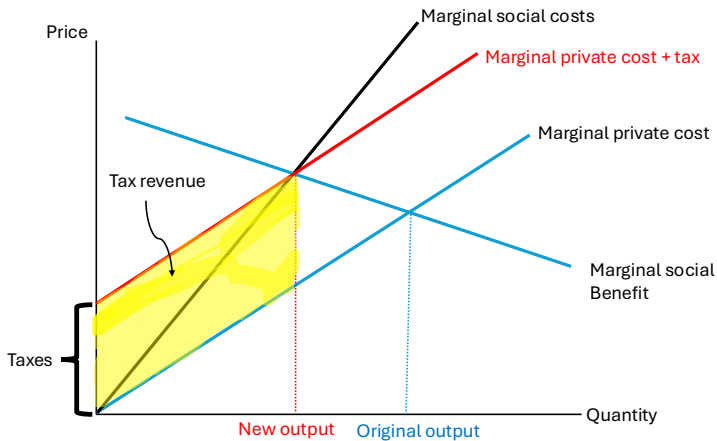
Carbon Tax



Carbon Tax



Carbon Tax



Carbon Tax

- ▶ carbon tax is intended to correct a negative externality (e.g., GHG emission)
- ▶ the tax is set equal to the marginal external cost of the negative externality, meaning the additional harm each unit of production or consumption imposes on society
- ▶ the socially optimal output occurs where marginal social cost equals marginal social benefit
- ▶ the marginal social cost reflects the sum of marginal private cost and marginal external cost
- ▶ the carbon tax internalizes the external cost, shifting the supply curve to match the marginal social cost curve

Carbon Tax

- ▶ when the carbon tax is applied, it raises the price to reflect the true social cost of each unit
- ▶ at the socially optimal quantity, the price consumers pay equals the marginal benefit of the good, which also matches the marginal social cost
- ▶ with carbon tax, the market equilibrium aligns with the point where $\text{price} = \text{marginal benefit} = \text{marginal social cost}$
- ▶ this condition corrects the market failure by ensuring the quantity produced is optimal from society's perspective

Carbon Tax Rate in Canada

- ▶ Canada's federal carbon tax is \$80 per tonne of carbon dioxide equivalent (CO_2e)
- ▶ this tax applies to fossil fuels such as gasoline, diesel, natural gas, and heating oil
- ▶ the scheduled rates are as follows:
 - ▶ 2019: \$20 per tonne CO_2e
 - ▶ 2020: \$30 per tonne CO_2e
 - ▶ 2021: \$40 per tonne CO_2e
 - ▶ 2022: \$50 per tonne CO_2e
 - ▶ 2023: \$65 per tonne CO_2e
 - ▶ 2024: \$80 per tonne CO_2e

The Canadian government has outlined plans to continue increasing the carbon tax by \$15 per tonne annually, aiming to reach \$170 per tonne CO_2e by 2030.

Question:

How does the government use the revenue collected from the carbon tax?

Answer:

- ▶ approximately 90% of the revenue is returned to individuals through the Canada Carbon Rebate
- ▶ the remaining 10% goes back to farmers, small-and medium-enterprises and Indigenous governments

Climate Change and Sustainability

Carbon tax is considered a component of sustainability as it aligns economic incentives with environmental goals by promoting reduced greenhouse gas emissions and encouraging clean energy adoption

How does it fit sustainability?

- ▶ **Environmental Protection:** The carbon tax discourages the use of fossil fuels by making carbon emissions costly. By putting a price on carbon, it helps lower pollution levels and mitigate climate change, directly supporting environmental sustainability
- ▶ **Resource Efficiency:** Carbon taxes encourage more efficient use of resources by motivating businesses and individuals to adopt energy-efficient practices and technologies

Climate Change and Sustainability cont'd

- ▶ **Economic Sustainability:** Revenue from carbon taxes can be invested in renewable energy projects, energy-efficient infrastructure, and public rebates, which contribute to economic resilience and a green transition
- ▶ **Social Equity:** When combined with measures like rebates (e.g., Canada's Carbon Rebate), carbon taxes can be structured to protect low- and middle-income households from financial burdens, supporting social sustainability by ensuring fair distribution of costs

Therefore, a carbon tax is a policy tool that supports sustainability by addressing economic, environmental, and social dimensions in the effort to combat climate change

Case Studies - In-Class Group Exercise

Form groups of 3-4 students and select one of the following case studies to research and discuss with your group members.

1. British Columbia's Carbon Tax
2. Ontario's Green Energy Act (GEA)
3. Alberta's Oil Sands and Climate Initiatives
4. Indigenous-Led Conservation in the Great Bear Rainforest

Critical Thinking Question:

Given Canada's commitment to increasing the carbon tax to \$170 per tonne by 2030, evaluate the potential economic and social impacts of this policy on different sectors (e.g., households, businesses, and energy-intensive industries). Consider both positive and negative effects. Do you think the revenue-recycling strategy through rebates is effective in addressing these impacts, or should additional measures be taken to ensure fairness and effectiveness in achieving Canada's climate goals?

Answer to Critical Thinking Question:

Will be discussed during next Lecture

References

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- ▶ Reed, G., Fox, S., Littlechild, D., McGregor, D., Lewis, D., Popp, J., Wray, K., Kassi, N., Ruben, R., Morales, S. and Lonsdale, S. (2024). For Our Future: Indigenous Resilience Report. Ottawa, Ontario.